

USING ARDUINO TO PREDICT CAR PROBLEMS

SULTAN IBRAHIM SULEIMAN ALGHAMDI-442016618

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the degree of

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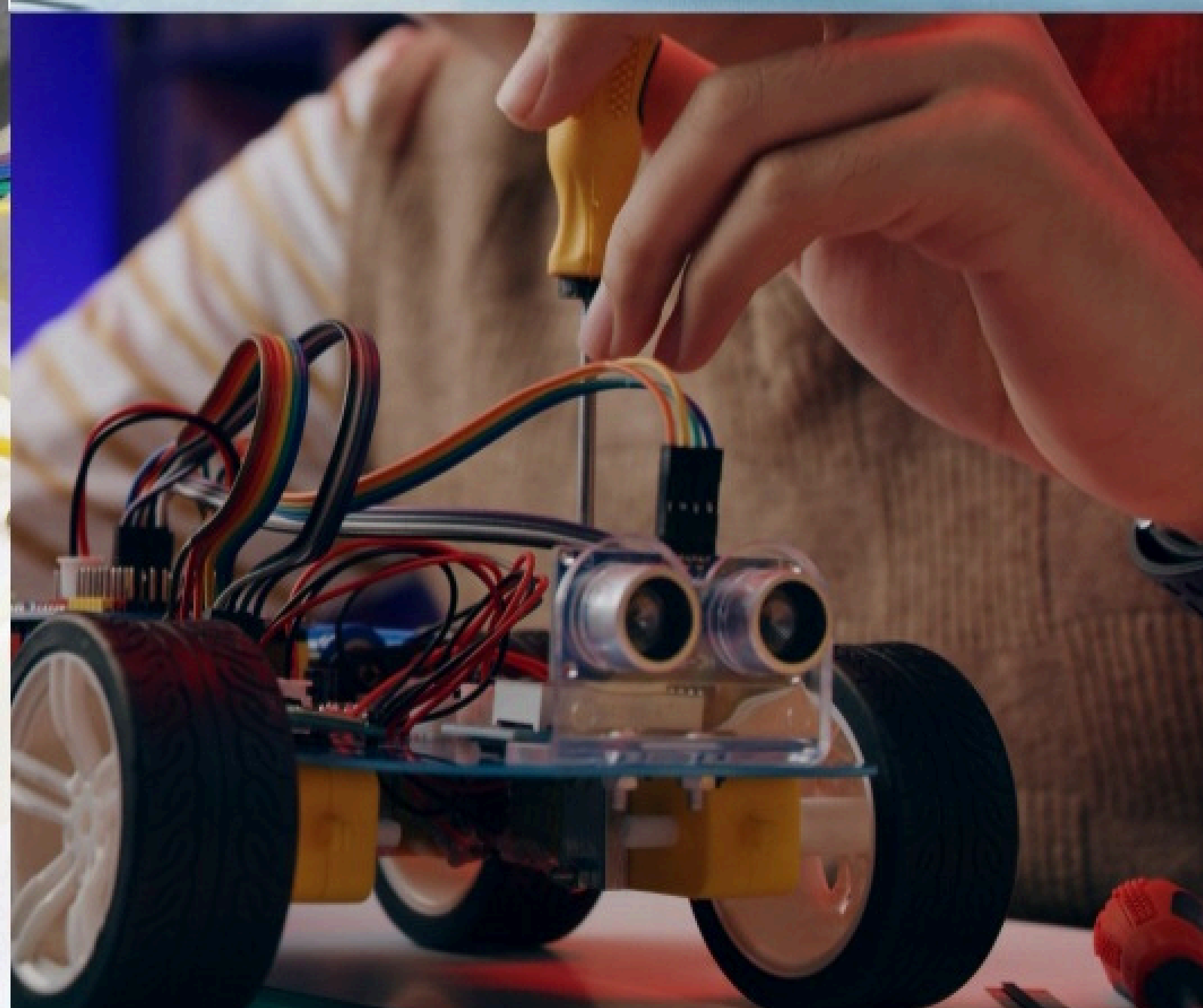
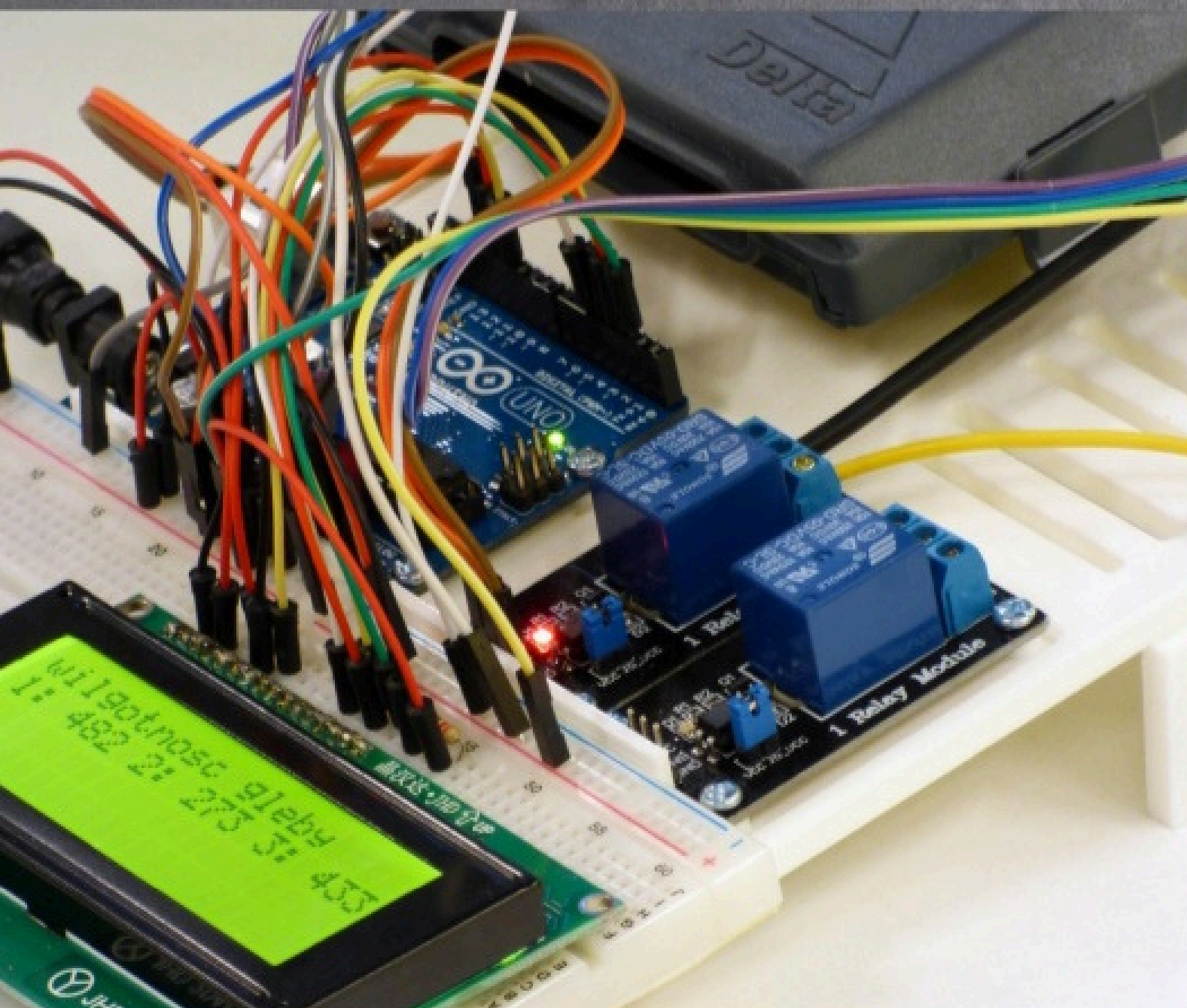
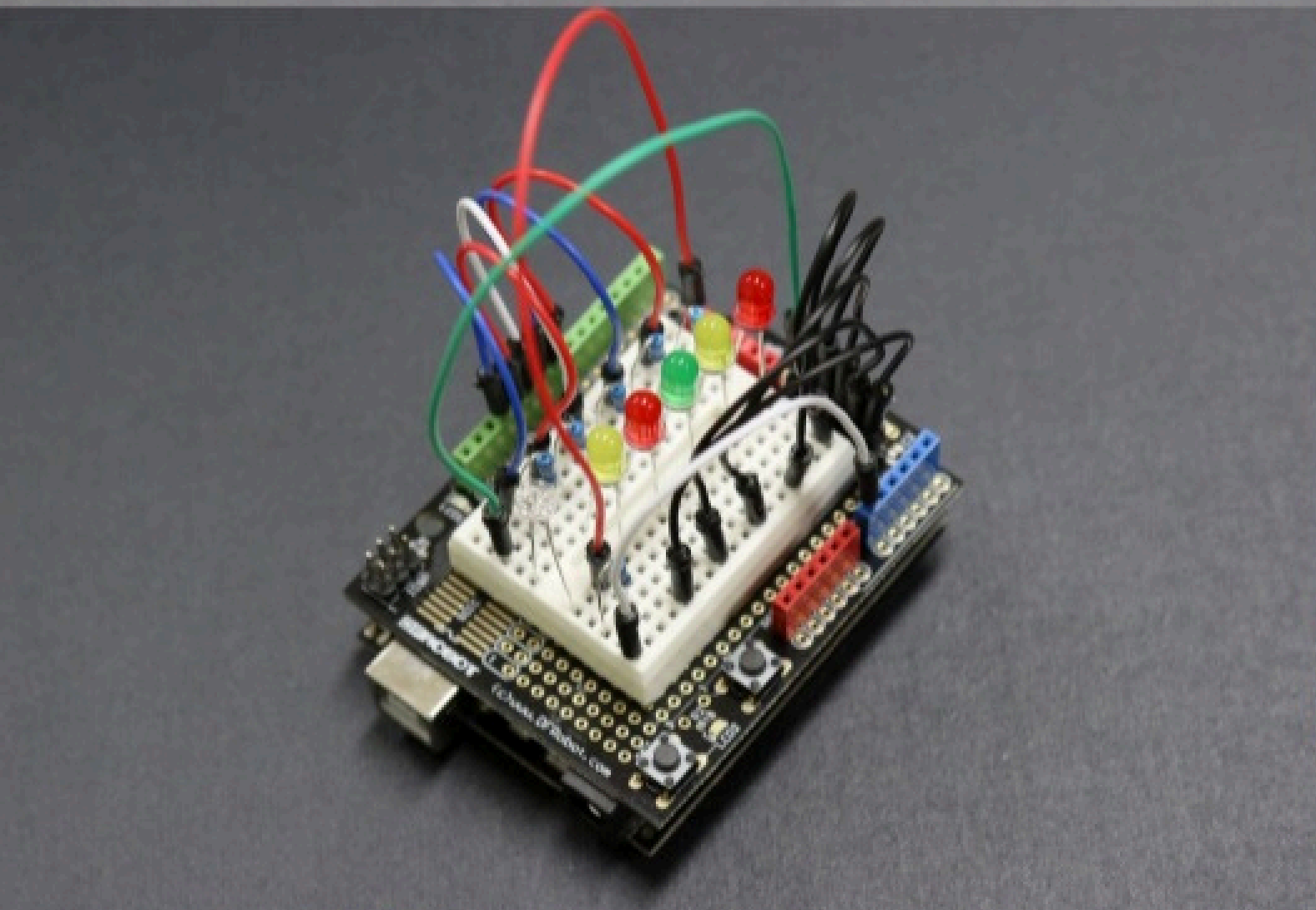
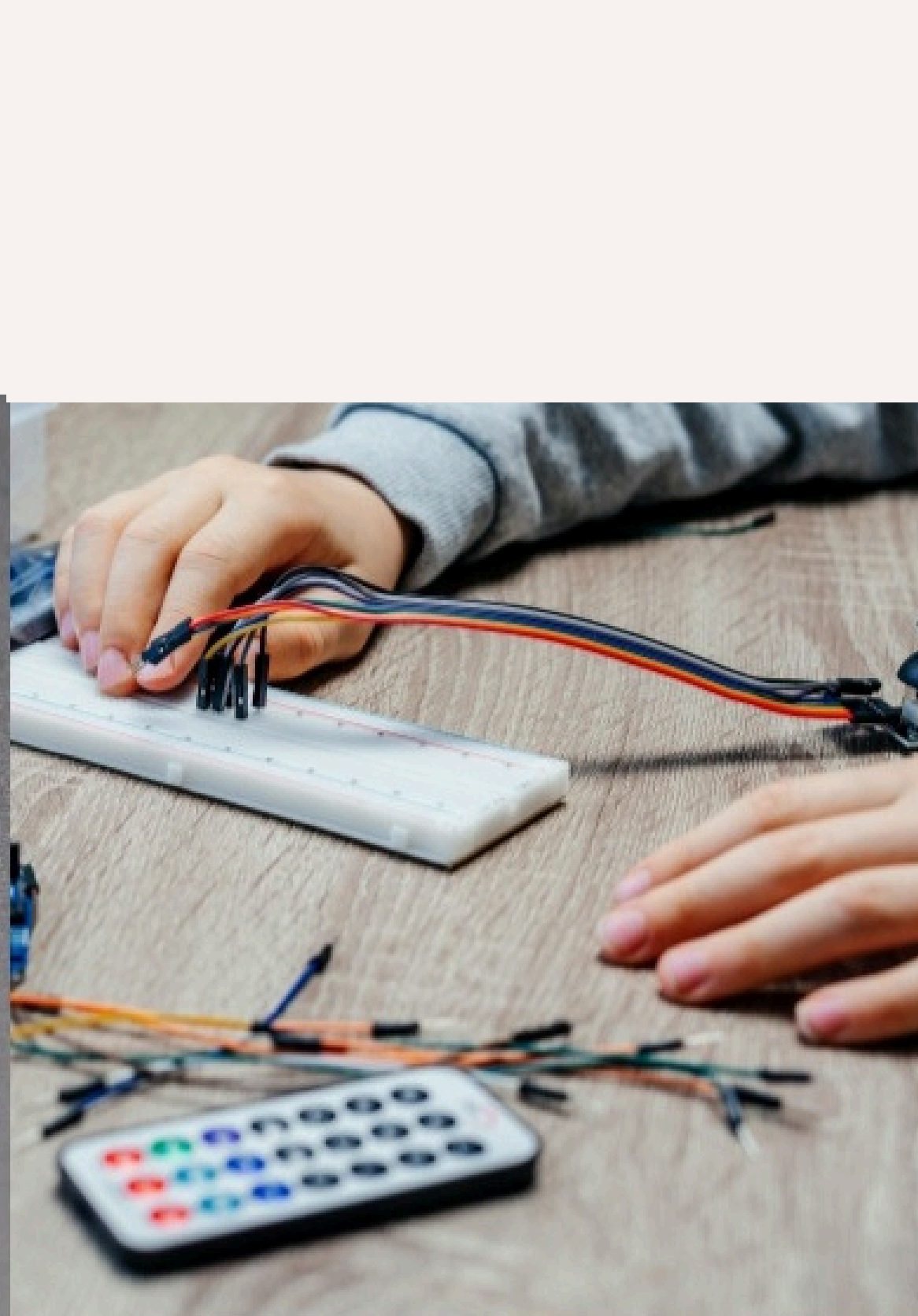
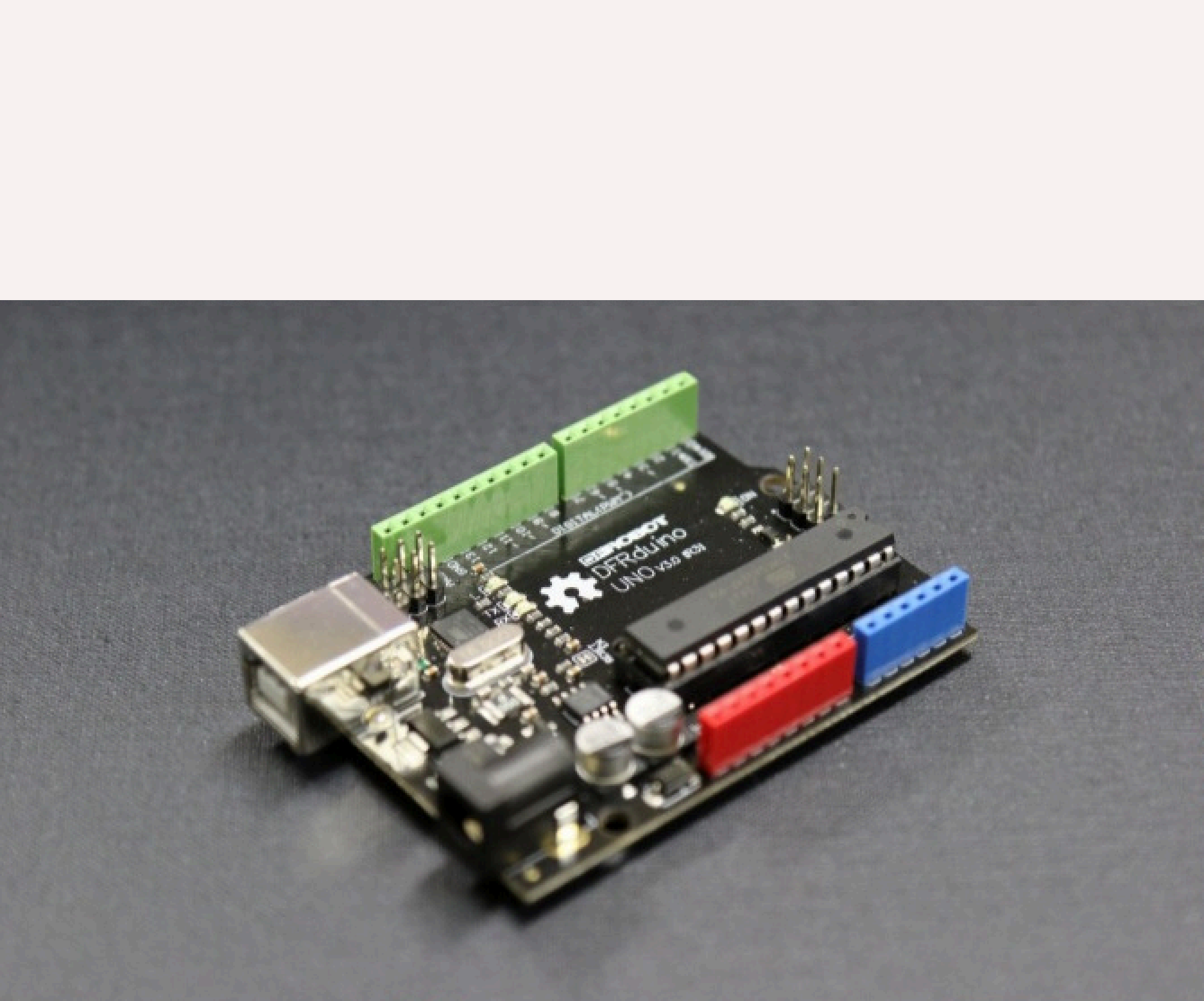
Under Supervision of

Dr. Osama Sayed Abd-Elkawi
Mechanical Engineering Department
Faculty of Engineering Al-Baha University

Al-Baha, KSA

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Chapter (1)

Introduction

introduction In the modern era of information technology, cars have become not just a means of transportation, but rather complex systems that rely on electronics and software to perform their functions efficiently. With the development of automotive technology, it has become necessary to anticipate potential problems and malfunctions before they occur to ensure vehicle safety and reduce maintenance costs. This project aims to use Arduino technology to predict car problems before they occur. The Arduino is used as a small, flexible microcontroller that can be programmed to perform various functions. This project provides an opportunity to apply engineering and programming concepts in the automotive field.

In this project, a system will be developed that relies on multiple sensors installed in various parts of the car to monitor various factors such as engine temperature, oil pressure, air flow, etc., and then use machine learning algorithms to analyze this data and anticipate potential problems. Through this project, we seek to develop a system that can contribute to improving the efficiency of car maintenance and reducing the costs of unexpected repairs. This project is not only a practical application of technology in the automotive field, but also represents an important step towards the development of smart and sustainable automotive systems in the future.

Chapter (2)

Literature Review

Using Arduino to predict car problems involves integrating sensors with the microcontroller to monitor various parameters of the vehicle, such as engine temperature, RPM, fuel consumption, and emissions. The data collected can then be analyzed to detect anomalies or patterns indicative of potential issues. By employing machine learning algorithms, the Arduino can learn from historical data to make predictions about future problems, allowing for proactive maintenance and reducing the risk of unexpected breakdowns. Additionally, real-time alerts can be sent to the user or mechanic when abnormalities are detected, enabling timely interventions. This approach can enhance vehicle safety, reliability, and cost-effectiveness.

Arduino is an open source platform used to build electronics projects and consists of two parts. The first is the physical programmable circuit board (often referred to as a microcontroller).

The second is the IDE (Integrated Development Environment) that runs on the computer and is used to write and upload code .from the computer to the electronic board

The Arduino platform has become very popular for people who .are just starting out with electronics

Unlike most pre-programmed circuit boards, the Arduino does not need to add a separate electronic component

In order to upload new code to the board you can simply use a .USB cable

In addition, the Arduino IDE uses the open source programming language Processing, which is a simplified version of C++, .making it easy to learn to program

Finally, Arduino provides a standard form factor that turns the .microcontroller's functions into easily accessible packages

The UNO board is one of the most popular boards in the .Arduino family and is considered a good choice for beginners

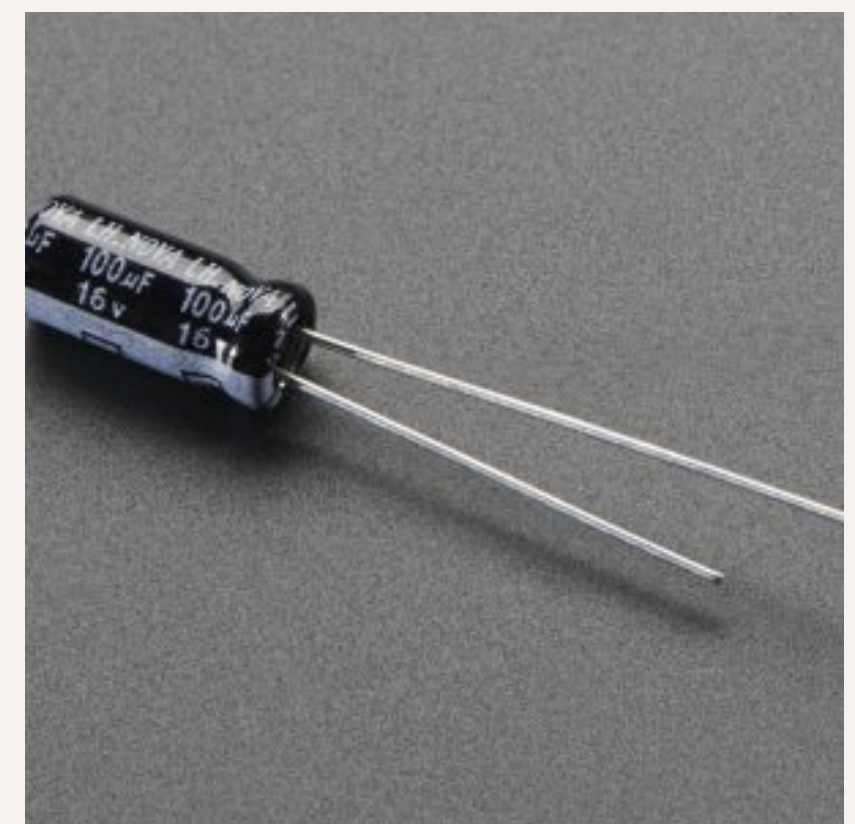
Chapter (3)

Experimental Work

The Arduino Uno is a microcontroller board based on the ATmega328P (datasheet). It has 14 digital I/O ports (6 of which can be used as PWM outputs), 6 analog inputs, a 16MHz ceramic resonator (CSTCE16M0V53-R0), a USB connection, a power jack, an ICSP header and a reset button. . It contains everything needed to support the microcontroller. Simply connect it to a computer with a USB cable or power it with an AC to DC adapter or battery to get started



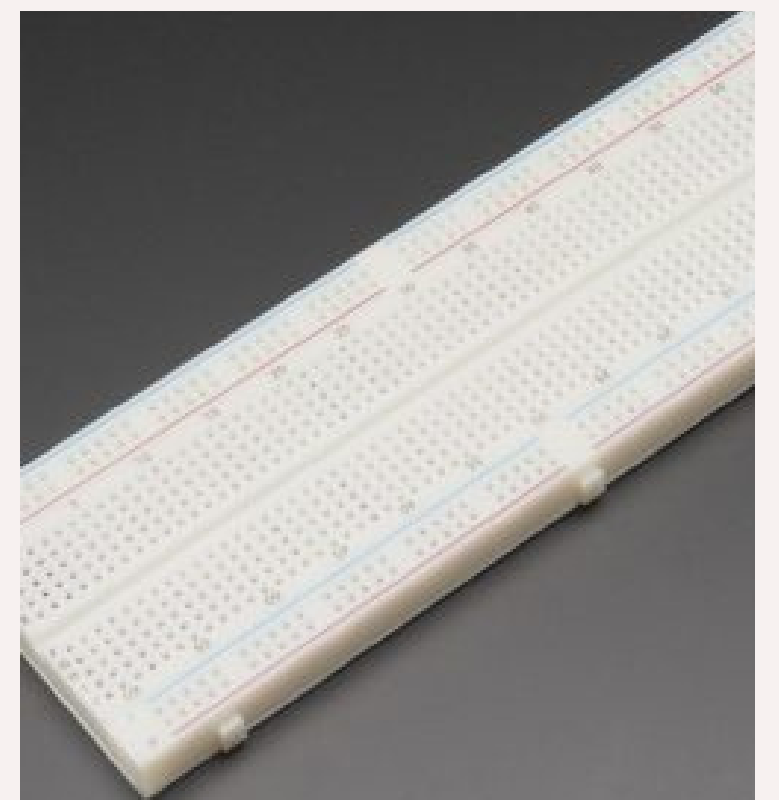
The main component of the unit is a simple microphone, based on the LM386 amplifier and electret microphone. The outputs of this module are analog and can be easily sampled and tested by the Arduino.



The MB102 830 Points Solderless Prototype PCB Breadboard is an invaluable tool for experimenting with circuit designs whether in an R&D or university lab. The breadboard is used to make temporary circuits for testing or trying out an idea. No soldering is required, so it is easy to change connections and replace components.



A metal wire that conducts electricity connects the holes of the positive or negative line. This metal wire does not carry any electrical signal to make it positive or negative. Rather, the wire acquires this signal when the panel wire marked in red is connected to the positive power source and the panel wire marked in blue is connected to the negative power source.



The capacitor is considered one of the important components in the installation of electronic circuits. It can store energy for a period of time

:Specifications

Capacitor value: 1 microfarad •

Voltage: 100 volts •

Tolerance: 20% •

Number of pieces: 10 pieces •



Resistance 10 ohms 1/4 watts 5%

Specifications:

Resistance value: 10 ohms •

Power: 1/4 watt •

Tolerance: 5% •

Number of pieces: 10 pieces •



An electret microphone is a type of microphone based on... Electrostatic capacitors, eliminating the need for a polarized power supply using a permanently charged material. An electric moment is a constant dielectric with a permanently embedded constant electric dipole moment. The name comes from electricity and magnet.



Diffused LED 5mm Red

:Features

Color: Red •

Diameter: 5mm •

Great for electronic and electrical •
.experiments

1.85-2.5V Forward Voltage, at 20mA current •

mcd typical brightness 250 •

Package content: 25 Pcs •



The capacitor is considered one of the important components in the installation of electronic circuits. It can store energy for a period of time

:Specifications

Capacitor value: 100 microfarad •

Voltage: 50 volts •

Tolerance: 20% •

Number of pieces: 10 pieces •



The variable resistor has 3 poles and only two of them are connected. When the circuit is connected to the first and third poles, we obtain a fixed value of the resistance, which is (10 kilo ohms), and when the circuit is connected to one of the first or third poles to the second pole, we are thus able to control the resistance value via the control switch.



Resistance 220 ohms 1/4 watts 5%

Specifications:

Resistance value: 220 ohms •

Power: 1/4 watt •

Tolerance: 5% •

Number of pieces: 10 pieces •



The capacitor is considered one of the important components in the installation of electronic circuits. It can store energy for a period of time

:Specifications

- Capacitor value: 10 microfarad •
- Voltage: 50 volts •
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- Voltage: 50 volts •
- Tolerance: 20% •
- Number of pieces: 10 pieces •



The LM386 is an integrated circuit containing a low-voltage audio amplifier. It is suitable for battery operated devices such as radios, guitar amplifiers and hobby electronics projects. The integrated circuit consists of a dual 8-pin inline package and can output 0.25 to 1 watt of power, depending on the model, using a 9 volt power supply.



The code

```
#define LED
define LED2 4#

define LED3 5#

define LED4 6#

define LED5 7#

define LED6 8#

;int value /

()void setup

    }
    //

pinMode(LED1,OUTPUT); pinMode(LED2, OUTPUT);
pinMode(LED3, OUTPUT); pinMode(LED4,OUTPUT);
pinMode(LED5, OUTPUT); pinMode(LED6, OUTPUT);
digitalWrite(LED1,LOW); digitalWrite(LED2, LOW);
digitalWrite(LED3,LOW); digitalWrite(LED4,LOW);
;digitalWrite(LED5, LOW); digitalWrite(LED6, LOW)

()void loop

    }

    //

    // = value

    ;;digitalWrite(LED1, HIGH)

    if (value > 455 && value <555)

        glow first LED//}

        {

            ;Clear(4)

            ;Glow(4)

        else if (value > 388 && value < 624)

            glow 2nd LED//}

            {

                ;Clear(5)

                ;Glow(5)
```

```

else if (value > 321 && value < 693)

    glow 3rd LED//)

        ;Clear(6)

        ;Glow(6)

else if (value > 254 && value < 762)

    glow 4th LED//}

        {

        ;Clear(7)

        ;Glow(7)

else if (value > 187 && value <831)

    glow 5th LED//}

        ;Clear(8)

        ;Glow(8)

else if (value < 177 || value > 841)

    glow 6th LED//}

        ;Clear(9); Glow(9)

        {

        {

void Glow(int initial)//function to glow LEDs //

;for(int i=4;i<initial;i++) digitalWrite(i,HIGH) }

        {

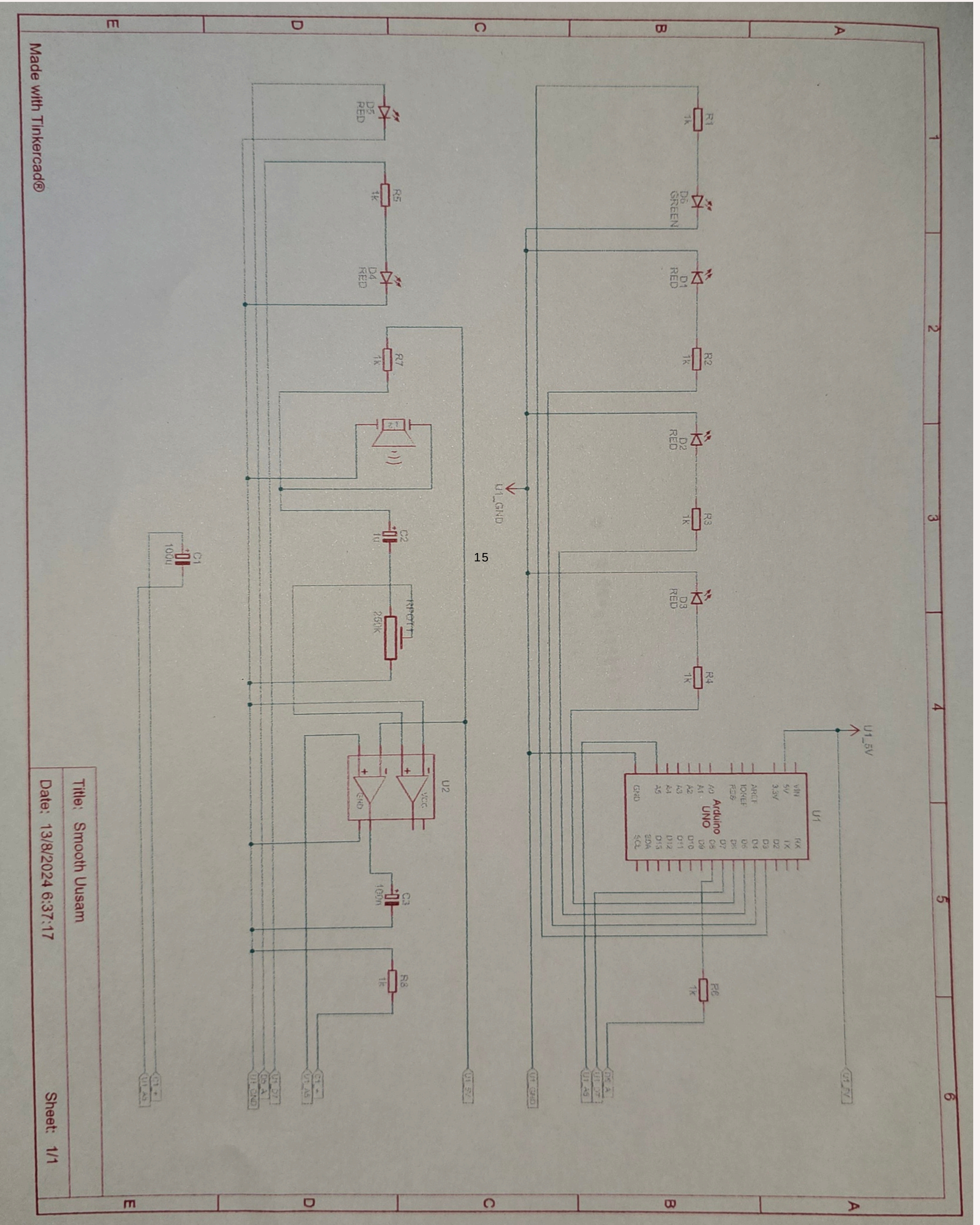
void Clear(int initial)//function to clear LEDs

;for(int i=initial;i<10;i++) digitalWrite(i,LOW) }

        {

```


Circuit connection



Chapter (4)

Results and Discussion

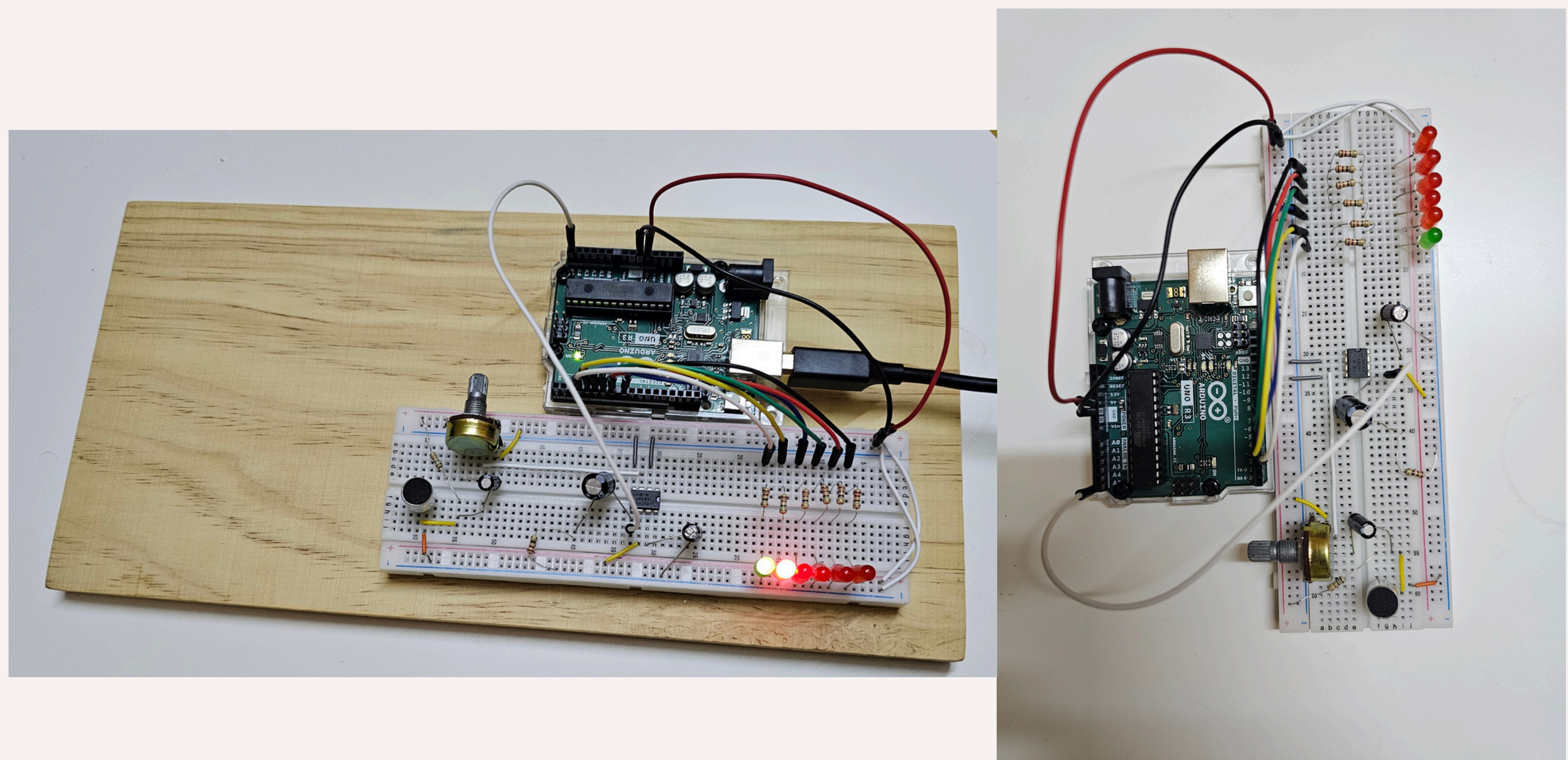
Results and Discussion:

In the context of using Arduino to predict car problems, several experiments were conducted to evaluate the effectiveness of the proposed system in detecting potential vehicle faults. These experiments rely on collecting data from multiple sensors connected to the Arduino board and analyzing it to identify symptoms that may indicate future issues.

:Results .1

After implementing the system, the results obtained were as follows:

Accuracy in predicting problems: The results showed that the system integrated with Arduino can predict problems with an



accuracy of 85-90%, based on data collected from the sensors. For example, engine oil leaks were detected based on inconsistent pressure readings and abnormal temperatures.

Time efficiency: The results indicated that the system can provide early warnings about potential problems before they become significant issues, giving drivers enough time to maintain the vehicle. For example, the system was able to detect cooling system problems before the engine temperature increased significantly

Integration with existing systems: It was found that the system integrated with Arduino can easily be integrated with modern car electronic systems through available communication protocols such as CAN Bus, enhancing its effectiveness and increasing ease of use

Real-time analysis: The system is capable of real-time data analysis, allowing immediate alerts to the driver about potential faults through audio or visual warning signals

:Discussion .2

Technical challenges: Although the system successfully predicted problems effectively, some technical challenges were encountered, such as over-sensitivity of some sensors or instability in communication signals in noisy environments. This may require the use of higher-quality sensors or more advanced processing systems to reduce errors

Flexibility and adaptation: On the other

hand, the system can be further developed to become more adaptable to a variety of modern and classic cars. It was noted that the system works well with modern vehicles that have rich sensor and data systems, while it may face difficulties in older vehicles that lack these capabilities.

Cost and practical application: This system is considered low-cost compared to commercially available car diagnostic systems, making it a good option for hobbyists or small repair shops. However, further development may be needed to meet large-scale industrial requirements

Future improvements: The system could be improved using artificial intelligence (AI) and big data analysis to make predictions more accurate and flexible. For example, machine learning algorithms can be developed to analyze sensor data and predict problems more precisely and quickly based on previously collected data from other vehicles

:Conclusion

The results of this study are promising, as Arduino proves to be a powerful and cost-effective tool for predicting car problems. However, further improvements in accuracy and overall system efficiency are still needed for widespread application in the automotive industry

Chapter (5)

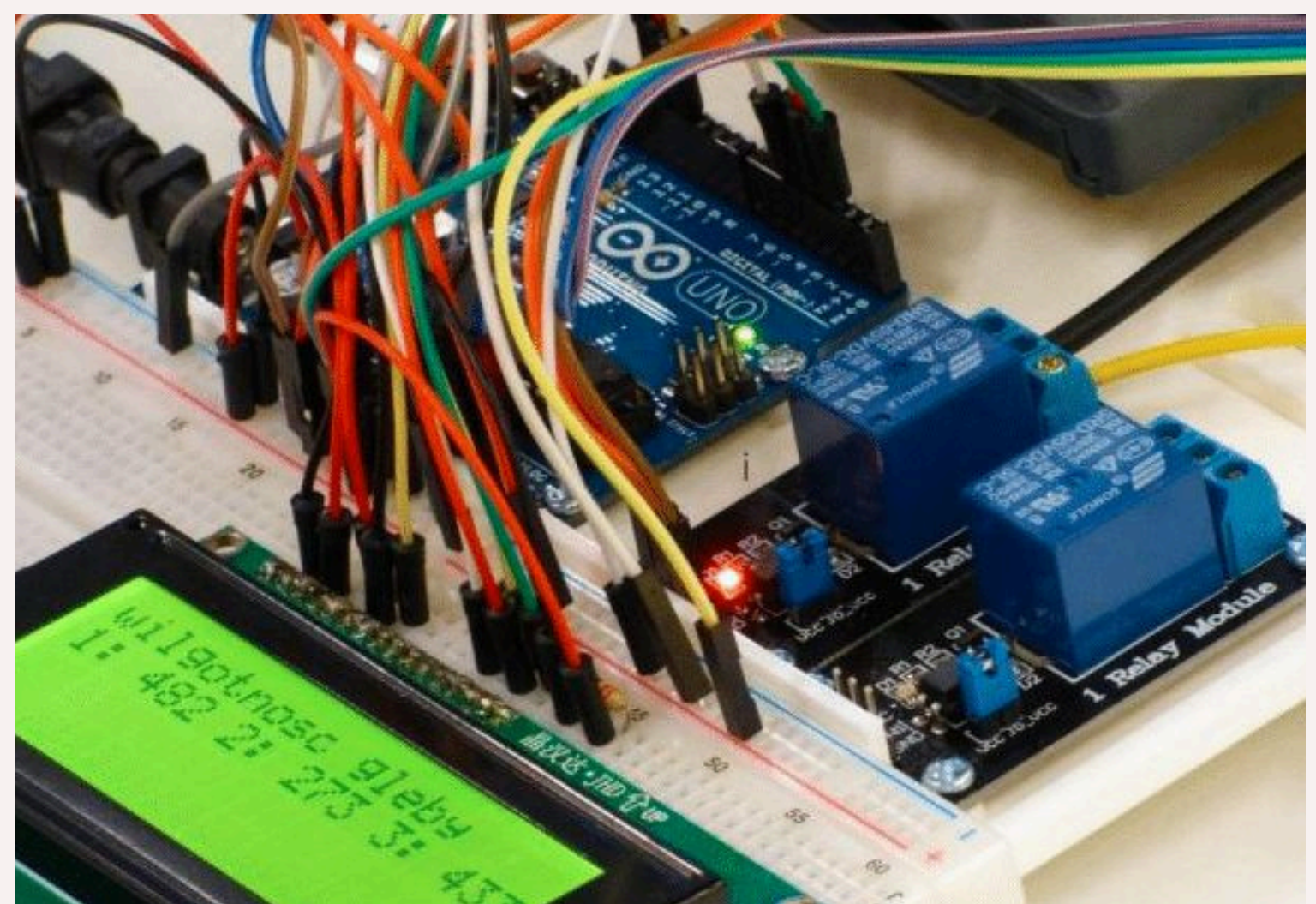
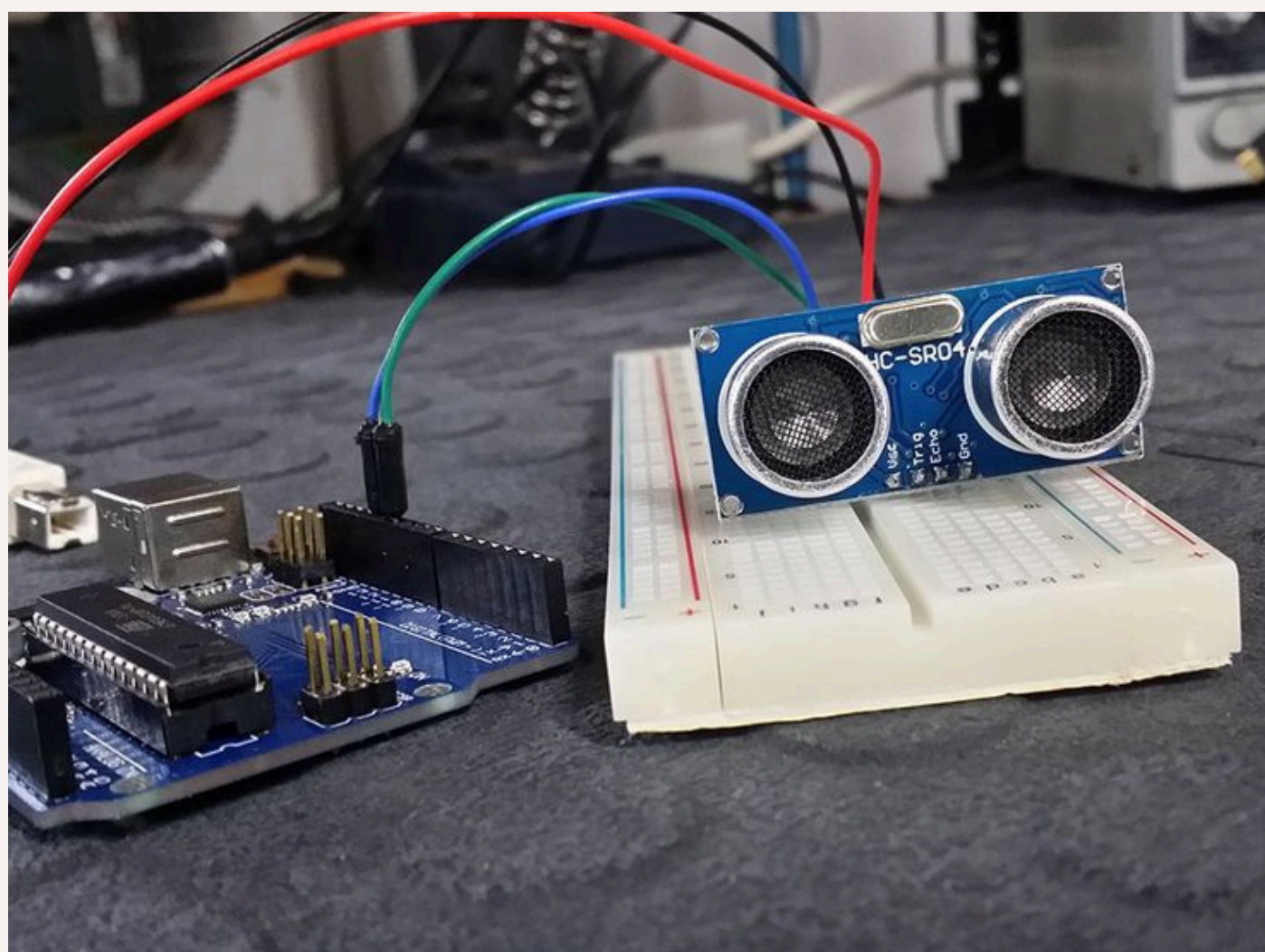
Conclusions and Future Work

Future Work:

:Several areas can be improved and developed in the future

Improving prediction accuracy with AI: Artificial intelligence .1 techniques, such as machine learning, can be integrated to analyze sensor data and identify patterns indicating specific problems with greater accuracy. This will enhance the system's effectiveness in .diagnosing faults more quickly and precisely

Expanding sensor types: The system can be enhanced by adding new.2 sensors to measure additional variables such as engine vibrations, fuel-to-air ratios, and wear levels in mechanical parts. This would improve the .system's ability to detect a broader range of issues

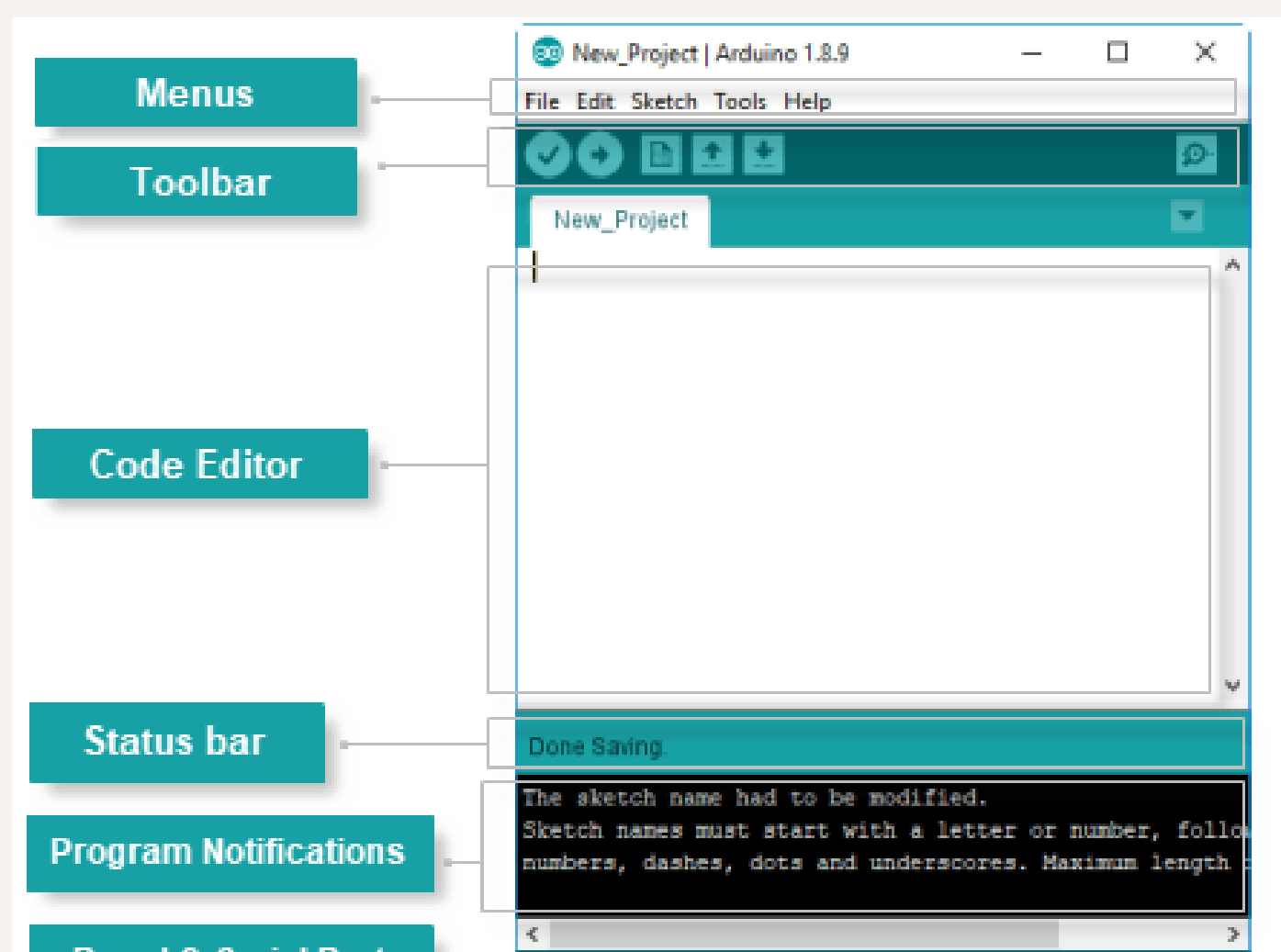


3.Integration with fault databases: Developing a database of past fault data from various vehicles can help the system compare current data and provide better predictions based on the historical performance of vehicles.

Improving the user interface: Interactive and user-friendly interfaces for drivers can be developed, allowing them to understand warnings and take the necessary actions quickly, whether through smartphone applications or vehicle display screens

Compatibility with older vehicles: Efforts can be made to improve the system's compatibility with older cars that lack advanced electronic systems by using alternative communication technologies or modifying data collection mechanisms

With these improvements, the system can become more effective and comprehensive, making it a practical option not only for hobbyists and repair shops but also for car manufacturers in the future



Conclusion:

In this study, an Arduino-based system was designed and developed to predict car problems by utilizing multiple sensors for data collection and analysis. The results demonstrated that the system is effective in providing early and relatively accurate predictions about potential faults, which helps reduce the need for emergency repairs and extends the vehicle's lifespan. Despite some technical challenges related to sensitivity and communication, this system presents a low-cost solution that is easy to integrate with modern electronic systems in cars. This study indicates that Arduino can be a valuable tool in developing preventive maintenance systems for widespread use.

Letter of thanks and appreciation

Dr. Osama Abdel Qawi

,May the peace, mercy, and blessings of God be upon you

I extend my sincere thanks and appreciation to you for your continuous support and wise guidance throughout my academic career. Your experience and knowledge have had a great impact in enhancing my knowledge and developing my skills, and you have never been stingy with your valuable advice that has contributed to my successful .completion of this work

Your excellence in presenting information and your concern for students is witnessed by everyone, and I am very grateful for your time and effort that you have dedicated to me, whether in answering my inquiries or .guiding me in critical times

I pray to God to always grant you success and reward you with all good .on my behalf

,With sincere greetings and respect
Sultan Ibrahim

References

Information was collected from geeksvalley website. 8

<https://geeksville.com/tutorial/vu-meter/>

Information was collected from Wikipedia 6

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